

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Long-Term Safety and Tolerability of Faricimab in Participants With Neovascular Age-Related Macular Degeneration **Official Title:** A Multicenter, Open-Label Extension Study to Evaluate the Long-Term Safety and Tolerability of Faricimab in Patients With Neovascular Age-Related Macular Degeneration (AVONELLE-X) **Short Title/Acronym:** AVONELLE-X **Protocol Number:** GR42691 **NCT Number:** NCT04777201 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Faricimab (Vabysmo) [ATC Code: S01LA09] **Phase of Development:** Phase III Extension Study **Trial Status:** COMPLETED **Medical Monitor:** Roche Clinical Trial Leader

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of faricimab 6 mg administered by intravitreal injection at a personalized treatment interval (PTI) in patients with neovascular age-related macular degeneration (nAMD). **Secondary Objectives:** To evaluate long-term efficacy, durability of visual improvements, maintenance of Best Corrected Visual Acuity (BCVA), anatomic improvements (such as resolution of retinal fluid and polypoidal lesions), and to determine the proportion of patients achieving extended dosing intervals (e.g., up to three to four months or more). **Substudy Objective:** To evaluate the impact of intravitreal faricimab on the health of the corneal endothelial cells in the study eyes to fulfill a U.S. Food and Drug Administration (FDA) post-marketing requirement, with the fellow eyes of the same participants serving as controls.

2.2 Background & Rationale Neovascular age-related macular degeneration (nAMD) can cause severe vision loss driven by the destabilization of blood vessels through pathways involving angiopoietin-2 (Ang-2) and vascular endothelial growth factor-A (VEGF-A). Faricimab (Vabysmo) is a bispecific antibody designed to target and inhibit both of these signaling pathways, stabilizing blood vessels and reducing inflammation. Following the two-year Phase III TENAYA and LUCERNE trials, AVONELLE-X was initiated as a two-year extension study to confirm faricimab's long-term safety, tolerability, and the viability of a treat-and-extend regimen. This approach helps address unmet needs in vision care by reducing the frequency of intravitreal injections while supporting long-term disease management.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Open-label Extension) **Control Type:** Single-arm for main study (All patients receive active treatment in the extension phase); Fellow-eye control for the substudy. **Blinding:** Open-label **Randomization:** N/A (Participants rolled over directly from parent trials)

3.2 Planned Enrollment & Duration Target **\$\$\$:** 1,036 participants
Number of Sites: Multicenter, Global **Estimated Study Start:** March 2021 **Study Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with a confirmed diagnosis of neovascular age-related macular degeneration (nAMD). 2. Successful enrollment in, and completion of, the end-of-study visit for one of the Phase III parent studies: GR40306 (TENAYA; NCT03823287) or GR40844 (LUCERNE; NCT03823300), without study or study drug discontinuation. 3. Provided signed informed consent to participate in this long-term extension study. 4. For women of childbearing potential: agreement to remain abstinent or use contraception with a failure rate of < 1% per year during the treatment period and for 3 months after the final dose of faricimab, and agreement to refrain from donating eggs during the same period. 5. **Substudy Inclusion:** Must sign an informed consent for the substudy, be able to participate for at least 48 weeks with at least the first visit while enrolled in the main study, and have a difference of <10% in corneal endothelial cell density at screening between the two eyes as measured by specular microscopy.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intending to become pregnant during the study or within 3 months after the final dose of faricimab. 2. Presence of other ocular diseases, metabolic dysfunction, or clinical laboratory finding that contraindicates the use of faricimab, renders the patient at high risk of complications, or interferes with study assessments. 3. History of a severe allergic reaction or anaphylactic reaction to a biologic agent or known hypersensitivity to any component of the faricimab injections, dilating drops, anesthetics, or antimicrobial preparations used. 4. Requirement for continuous use of prohibited therapy. 5. **Substudy Exclusions (Study or Fellow Eye):** - Prior/current administration of faricimab or brolucizumab in the fellow (non-study) eye. - Corneal endothelial cell density ≤ 1500 cells/mm² at screening. - Fuchs endothelial corneal dystrophy Grade ≥ 2 . - Previous ocular trauma or corneal endothelial cell damage. - Any condition precluding analyzable specular microscopy images. - Active or history of corneal edema, corneal dystrophies, iridocorneal endothelial syndrome, pseudoexfoliation syndrome, or herpetic keratitis/kerato-uveitis. - Intraocular laser therapy (e.g., SLT, YAG, prophylactic

peripheral iridotomy) within 1 year, or YAG capsulotomy within 3 months of screening. - Prior vitrectomy, pars plana vitrectomy, submacular surgery, or other surgical intervention for AMD. - Previous intraocular device implantation (excluding intraocular lenses), cataract surgery within 6 months or planned, or history of corneal transplantation. - History of glaucoma-filtering surgery, tube shunts, or microinvasive glaucoma surgery (other prior glaucoma surgeries allowed if >6 months before screening). - Administration of topical Rho kinase inhibitors within 1 month prior to screening. - Contact lens wear in either eye within 2 months of screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Faricimab 6 mg administered at a personalized treatment interval (PTI) based on a "treat-and-extend" concept. The time between treatments is adjusted dynamically based on the patient's retinal fluid levels and visual acuity (often extended to every 3 to 4 months). **Route of Administration:** Intravitreal injection. **Duration of Treatment:** 2 years (104 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard ocular care and medications for comorbidities not affecting the eye. **Prohibited:** Administration of any other anti-VEGF therapies (e.g., aflibercept) or investigational treatments in the study eye during the extension period. Continuous use of any medications indicated as prohibited therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening / Baseline	Day 0 Rollover from end-of-study visit of parent trials (TENAYA/LUCERNE), Baseline BCVA, OCT imaging, First Extension Dose
2	Interim	Per PTI (e.g., every 4 to 16+ weeks) Ocular safety exams, BCVA assessment, Retinal imaging (OCT), Faricimab injection based on treat-and-extend criteria
3	End of Study	Year 2 (Week 104) Final ocular exam, final BCVA and imaging, safety follow-up, study exit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Incidence and Severity of Ocular Adverse Events in the Study Eye: Severity graded according to AE severity grading scale (mild, moderate, severe). Includes monitoring for Adverse Events of Special Interest (AESI). **2. Incidence of Ocular Adverse Events in the Fellow Eye:** Assessing seriousness, severity, and causality for AEs in the non-study eye. **3. Incidence and Severity of Non-Ocular (Systemic) Adverse Events.**

Note on AESIs: AESIs include sight-threatening AEs that cause a visual acuity (VA) score drop of ≥ 30 letters lasting >1 hour, severe intraocular inflammation (IOI), or AEs requiring surgical/medical intervention to prevent permanent sight loss; suspected transmission of an infectious agent; and potential drug-induced liver injury (elevated ALT/AST combined with elevated bilirubin or clinical jaundice, defined by Hy's Law).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in Best Corrected Visual Acuity (BCVA) measured by ETDRS letters. 2. Anatomic improvements, including change in central subfield thickness (CST) and resolution of retinal fluid or polypoidal lesions. 3. Proportion of patients assigned to extended dosing intervals (e.g., Q12W, Q16W, or Q20W) at the end of the study. 4. **Substudy Endpoint:** Percent Change in Corneal Endothelial Cell Density From Baseline at Week 24 in the Study Eye as Compared With the Fellow Eye.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of intraocular inflammation, increased intraocular pressure, sight-threatening AEs, and systemic AEs at every study visit. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External independent data monitoring committee assigned to periodically review accumulated safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety-evaluable population (all patients who received at least one dose of faricimab in the AVONELLE-X study). **Sample Size Justification:** Approximately 1,036 patients rolled over from the parent efficacy studies, representing an adequate sample size to detect rare safety signals over long-term exposure. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for continuous efficacy measures (e.g., BCVA), or observed cases analysis where appropriate.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Periodic onsite and remote risk-based monitoring to ensure protocol compliance and accurate reporting of ophthalmic imaging. **Audit Trail:** Complete electronic Case Report Form (eCRF) tracking with centralized reading center verification for all retinal and corneal imaging (specular microscopy).

11. Study Identifiers & Governance

Primary ID: GR42691 **Registry Number:** NCT04777201 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** AVONELLE-X **Official Regulatory Title:** A Multicenter, Open-Label Extension Study to Evaluate the Long-Term Safety and Tolerability of Faricimab in Patients With Neovascular Age-Related Macular Degeneration

12. Clinical Framework (Ophthalmology Specific)

Primary Condition: Neovascular Age-Related Macular Degeneration (nAMD) **Diagnosis Threshold:** Active choroidal neovascularization secondary to AMD established in parent trials **Medical Methodology:** Treat-and-extend / Personalized Treatment Interval (PTI) **Optimization Target:** Maintenance of visual acuity with the maximum tolerable interval between intravitreal injections (up to 4 months or longer)

13. Study Procedures & Methodology

Management Pathway: Standard intravitreal injection protocol **Education Protocol (HCP):** Training on PTI criteria for dose extension or reduction based on retinal fluid **Education Protocol (Patient):** Counseling on recognizing signs of ocular inflammation, endophthalmitis, or acute vision changes. **Quality Audit Mechanism:** Central reading centers for optical coherence tomography (OCT) and specular microscopy to ensure unbiased evaluation of anatomic and corneal endpoints.

14. Observation & Follow-Up Schedule

Total Duration: 2 Years (104 weeks) **Onsite Visit Cadence:** Individualized based on PTI (typically ranging from every 4 weeks to every 16+ weeks) **Remote Monitoring Frequency:** N/A (Ocular assessments require onsite evaluation) **Checklist Review Interval:** At every scheduled injection visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				